HW5 (due Feb. 18, 2010)

(1)

40 pts.

A paramagnet in one dimension can be modelled as a linear chain of $N+1$ spins. Each spin interacts with its neighbours in such a way that the energy is $U = n\epsilon$ where n is the number of domain walls separating regions of up spins from down as shown by a vertical line in the representation below.

$$\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow|\downarrow\downarrow\downarrow\downarrow|\uparrow\uparrow\uparrow\uparrow\uparrow|\downarrow\downarrow|\uparrow\uparrow\uparrow|\downarrow\downarrow\downarrow\downarrow\downarrow\downarrow\downarrow$$

How many ways can n domain walls be arranged? Calculate the entropy, $S(U)$, and hence show that the energy is related to the temperature as

$$U = \frac{N\epsilon}{\exp(\epsilon/k_{\mathrm{B}}T) + 1}.$$

Sketch the energy and the heat capacity as a function of temperature, paying particular attention to the asymptotic behaviour for low and for high temperatures.

(2)

20 pts.

The entropy of black body radiation is given by the formula $S = \frac{4}{3}\sigma V^{1/4}U^{3/4}$ where σ is a constant. Determine the temperature of the radiation and show that $PV = U/3$.

Note: $U = E$ here.

(3)

40 pts.

Consider a system of N non-interacting particles, each fixed in position and carrying a magnetic moment. The system is immersed in a magnetic induction field $\mathbf{B}$. Each particle may then exist in one of the two energy states: a spin-down state of energy $-\mu B$ and a spin-up state of energy μB. Treat the particles as being distinguishable.

Derive a formula for the entropy in terms of the number of particles in the upper state, n. Sketch $S(n)$. Find the value of n for which $S(n)$ is a maximum. What is the relation between S and the temperature? By treating the energy as a continuous variable, show that the system can have negative temperatures. If a system with a negative temperature makes thermal contact with one with a positive temperature, which way will the heat flow?

① N bonds and n domain walls

$$\Omega = \frac{N!}{(N-n)!\,n!} \qquad S = k_B \ln \Omega$$

$$\ln \Omega = \ln N! - \ln (N-n)! - \ln n!$$

Use Stirling's Approximation: $\ln \Omega = N \ln N - (N-n)\ln(N-n) - n \ln n$

$$\ln \Omega = N \ln \frac{N}{N-n} + n \ln \frac{N-n}{n}$$

Expressing n in terms of U so that we get $S(U)$

$n = U/\varepsilon$ Then $S(U) = k_B\left(N \ln \frac{N}{N - U/\varepsilon} + \frac{U}{\varepsilon} \ln \frac{N - U/\varepsilon}{U/\varepsilon}\right)$

$$S(U) = N k_B\left(\ln \frac{N\varepsilon}{N\varepsilon - U} + \frac{U}{N\varepsilon} \ln \frac{N\varepsilon - U}{U}\right)$$

Now use the relation $\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_N$

$$\frac{1}{T} = N k_B\left(\frac{N\varepsilon - U}{N\varepsilon} \cdot \frac{N\varepsilon}{(N\varepsilon - U)^2} + \frac{U}{N\varepsilon}\left(\frac{-1}{N\varepsilon - U} - \frac{1}{U}\right) + \frac{1}{N\varepsilon} \ln \frac{N\varepsilon - U}{U}\right)$$

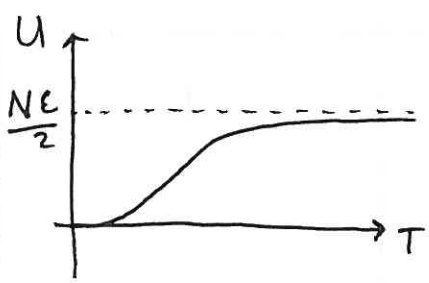
$$= N k_B\left(\frac{1}{N\varepsilon - U} - \frac{1}{N\varepsilon - U} + \frac{1}{N\varepsilon} \ln \frac{N\varepsilon - U}{U}\right)$$

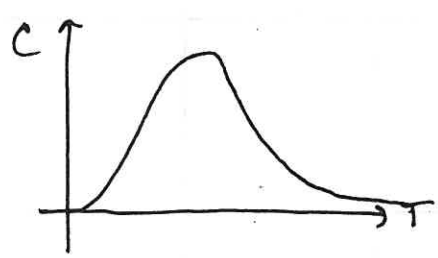
$$= N k_B\left(\frac{1}{N\varepsilon} \ln \frac{N\varepsilon - U}{U}\right) = \frac{k_B}{\varepsilon} \ln \frac{N\varepsilon - U}{U} = \frac{1}{T}$$

Now solve for U: $\frac{N\varepsilon - U}{U} = e^{\varepsilon/k_B T}$ $\quad N\varepsilon - U = U e^{\varepsilon/k_B T}$

So $U(T) = \dfrac{N\varepsilon}{e^{\varepsilon/k_B T} + 1}$ $\qquad U(T) = \begin{cases} \frac{N\varepsilon}{2} & T \to \infty \\ N\varepsilon e^{-\varepsilon/k_B T} & T \to 0 \end{cases}$

$C = \left(\frac{dU}{dT}\right)_N = \dfrac{N\varepsilon^2}{k_B T^2} \dfrac{e^{\varepsilon/k_B T}}{\left(e^{\varepsilon/k_B T} + 1\right)^2}$ $\qquad C(T) = \begin{cases} 0 & T \to \infty \\ 0 & T \to 0 \end{cases}$





② $S = \frac{4}{3}\sigma V^{1/4} U^{3/4}$

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_V = \frac{4}{3}\sigma V^{1/4}\frac{3}{4}U^{-1/4} = \sigma\left(\frac{V}{U}\right)^{1/4}$$

$$T = \frac{1}{\sigma}\left(\frac{U}{V}\right)^{1/4}$$

Show $PV = \frac{U}{3}$

$$dU = TdS - PdV \qquad dS = \frac{1}{T}dU + \frac{\sigma}{3}\left(\frac{U}{V}\right)^{3/4}dV$$

$$dU = T\frac{1}{T}dU + \frac{1}{\sigma}\left(\frac{U}{V}\right)^{1/4}\frac{\sigma}{3}\left(\frac{U}{V}\right)^{3/4}dV - PdV$$

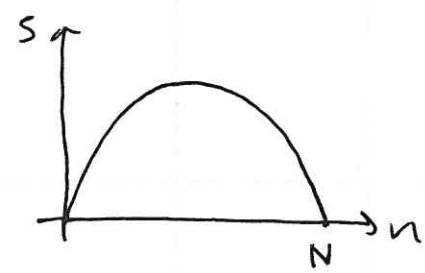
$$dU = dU + \frac{U}{3V}dV - PdV \qquad PdV = \frac{U}{3V}dV \qquad PV = \frac{U}{3}$$

③ If we add μB to each energy we transform the problem to something very similar to problem 1.

$$\varepsilon(\uparrow) = 2\mu B, \quad \varepsilon(\downarrow) = 0$$

If n is the number of up states we get the same solution for the entropy as problem 1.

$$S = k_B\left(N\ln\frac{N}{N-n} + n\ln\frac{N-n}{n}\right)$$



Find maximum $\frac{\partial S}{\partial n} = 0$

$$\frac{\partial S}{\partial n} = k_B\left(N\frac{1}{N-n} - \frac{N}{N-n} + \ln\frac{N-n}{n}\right) = 0$$

$$k_B\ln\left(\frac{N-n}{n}\right) = 0 \qquad \frac{N-n}{n} = 1 \qquad n = \frac{N}{2}$$

The total energy of the system is now $n\varepsilon$ $(\varepsilon = 2\mu B)$
Same result as problem 1

$$U(T) = \frac{N\varepsilon}{e^{\varepsilon/k_BT}+1} = \frac{2N\mu B}{e^{2\mu B/k_BT}+1}$$

③ cont..

Also like problem 1 $\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_N = \frac{k_B}{\varepsilon} \ln \frac{N\varepsilon - U}{U}$

or $T = \frac{\varepsilon}{k_B} \frac{1}{\ln\left(\frac{N\varepsilon - U}{U}\right)}$ if $\frac{N\varepsilon - U}{U} < 1$ T will be negative

$N\varepsilon - U < U$ $U > \frac{N\varepsilon}{2}$ If $U > N\mu B$ get negative temperature

~~This means over half of the spins are in the excited up state~~

The heat will flow from the negative temperature to the positive temperature. The negative temperature system has more energy $U > \frac{N\varepsilon}{2}$ than the positive temperature system. The asymptotic limit for positive temperature is $\frac{N\varepsilon}{2}$. The system will try to reach equilibrium by heat flow from negative system to positive